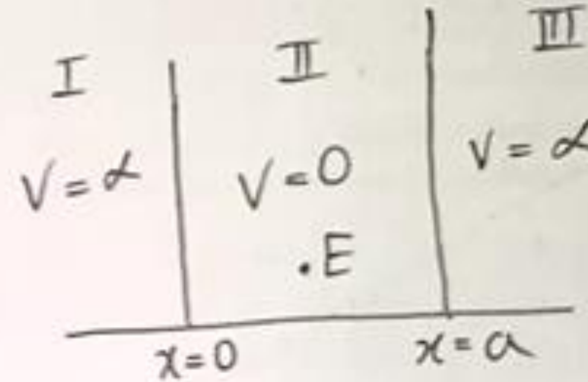
Unit-4
Lecture-2

Solution of stationary-state Schrodinger equation for one dimensional problems –

Square-well potential
&
Potential step

Application to one dimensional problem - square well potential



We know

$$\frac{d^2\psi}{dx^2}+\frac{2m}{\hbar^2}(E-V)\psi=0$$

for $0<x<a$

$$\frac{d^2\psi}{dx^2}+\frac{2mE}{\hbar^2}\psi=0$$

$$\frac{d^2\psi}{dx^2}+k^2\psi=0 \text{ — (i)} \quad \left(k^2=\frac{2mE}{\hbar^2}\right)$$

Sol^n of (i) $\underline{\psi = A\sin kx + B\cos kx}$

At $x=0 \Rightarrow 0=0+B$

$\Rightarrow B=0$

$\therefore \psi = A\sin kx$ — (ii)

At $x=a$, $\psi=0$

$0 = A\sin ka$

$ka = n\pi$

$k = \frac{n\pi}{a}$ — iii

Using normalization cond^n

$$\int_a^a A^2 \sin^2 kx\,dx = 1$$

$$A^2\int_0^a \sin^2 kx\,dx = 1$$

$$A^2 \times \left(\frac{x}{2}\right)_0^a = 1$$

$$A^2 = \frac{2}{a}$$

$$A = \sqrt{\frac{2}{a}}$$

$$\therefore \psi = \sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}x$$

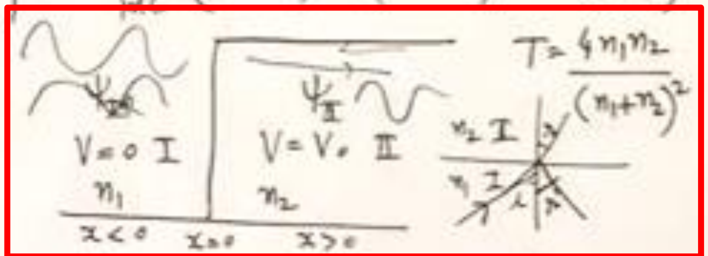
$V(x) = 0 \quad 0<x<a$

$V(x) = \infty \quad x>a, \; x<0$

Solution of stationary-state Schrodinger equation
for one dimensional problems –
Potential step

$$R=\left(\frac{p_1-p_2}{p_1+p_2}\right)^2=\left(\frac{n_1-n_2}{n_1+n_2}\right)^2 \quad T=\frac{4p_1p_2}{(p_1+p_2)^2} \quad R+T=1$$

Potential Step :-



$V=0$ for $x<0$

$=V_0$ for $x>0$

from Schrodinger equation

$$\frac{\partial^2\psi}{\partial x^2}+\frac{2m}{\hbar^2}(E-V)\psi=0 \quad -(1)$$

In Region I $V=0$

$$\frac{\partial^2\psi}{\partial x^2}+\frac{2mE}{\hbar^2}\psi=0 \quad -(2)$$

put $p_1^2=2mE$ $-(3)$ and

then $\psi_I=\underset{\psi_{inc}}{Ae^{ip_1x/\hbar}}+\underset{\psi_{ref}}{Be^{-ip_1x/\hbar}}$ $-(4)$

In Region II $V=V_0$ —

$$\frac{\partial^2\psi}{\partial x^2}+\frac{2m}{\hbar^2}(E-V_0)\psi=0 \quad -(5)$$

put $p_2^2=2m(E-V_0)$ $-(6)$

$$\psi_{II}=Ce^{\frac{ip_2x}{\hbar}}+De^{-ip_2x/\hbar} \quad -(7)$$

Here $D=0$

Applying Boundary Condition

$\psi_I=\psi_{II}$ $-(8)$

$\frac{\partial\psi_I}{\partial x}=\frac{\partial\psi_{II}}{\partial x}$ at $x=0$

$A+B=C$ $-(9)$

$p_1A-p_1B=p_2C$ $-(10)$

$$\frac{C}{A}=\left(\frac{2p_1}{p_1+p_2}\right) \quad -(11)$$

and $\frac{B}{A}=\left(\frac{p_1-p_2}{p_1+p_2}\right)$ $-(12)$

Here $B=$ Amplitude of Reflected Wave

$A=$ " " Incident "

$C=$ " " Transmitted Wave

Case I. If $E>V_0$ p_2 is real, then

$$(S)_I=\frac{\hbar}{2im}\left(\psi^*\nabla\psi-\psi\nabla\psi^*\right)$$

$$=\frac{p_1}{m}\left[|A|^2-|B|^2\right] \quad -(13)$$

Here $\frac{p_1}{m}|A|^2=$ probability Current of Incident beam S_I

$\frac{p_1}{m}|B|^2=$ " " " Reflected " S_R

$S_{II}=\frac{p_2}{m}|C|^2=$ " " " Transmitted beam S_T

$$R=\frac{S_R(\text{Ref P.C.D})}{S_I(\text{In. P.C.D})}=\frac{\frac{p_1}{m}|B|^2}{\frac{p_1}{m}|A|^2}=\left(\frac{p_1-p_2}{p_1+p_2}\right)^2 \quad -(14)$$

using eq (12)

$$T=\frac{S_T(\text{T.P.C.D})}{S_I(\text{I.P.C.D})}=\frac{\frac{p_2}{m}|C|^2}{\frac{p_1}{m}|A|^2}=\frac{p_2}{p_1}\left(\frac{2p_1}{p_1+p_2}\right)^2=\frac{4p_1p_2}{(p_1+p_2)^2} \quad -(15)$$

eq (11)

$p_1=mv_1=\frac{mc}{n_1}$ and $p_2=mv_2=mc/n_2$

$p_1=\sqrt{2mE}$

$p_2=\sqrt{2m(E-V_0)}$

$$\boxed{R=\left(\frac{n_2-n_1}{n_1+n_2}\right)^2 \text{ and } T=\frac{4n_1n_2}{(n_1+n_2)^2}}$$

$R+T=1$

$n_1=1.0$, $n_2=1.5$